Preprocessing for BFRB gesture detection

SC4000 group project · Kaggle “CMI – Detect Behavior with Sensor Data” · preprocessing workstream · 13 October 2026

In one paragraph. We rebuilt the preprocessing so that one function, `preprocess()`, cleans a single recording, corrects how the watch was worn and computes 117 features per time step, with exactly the same code in training and inside the Kaggle prediction call. Every physical assumption was checked on the real data before it was used, and every step was measured with the same model on the same five subject folds. The final pipeline scores **0.815** with all sensors and **0.740** with the motion sensor alone, against 0.750 and 0.663 for raw data: **+0.065** and **+0.078**.

Headline results

	All sensors	IMU only (≈ half of the hidden test set)
Raw data, missing → 0 (row A)	0.750 ± 0.012	0.663 ± 0.015
Final pipeline (row H)	0.815 ± 0.013	0.740 ± 0.016
Improvement	+0.065	+0.078

Score = ½ binary F1 (BFRB vs not) + ½ macro F1 over the 8 BFRB gestures plus one “non-target” class, the competition metric. Mean ± standard deviation over 5 folds split by person (StratifiedGroupKFold, seed 42). The measuring model is a fixed LightGBM classifier on per-recording summary statistics; it is a yardstick for the preprocessing, not our final model.

What the evidence says, in short

- **Biggest single wins:** ToF handling (treating “nothing detected” as far, plus 3×3 regions of each depth image), statistics from the end of the recording where the gesture happens, and fixing how the device was worn.
- **Left-handed and upside-down wearers** are real and correctable: after the corrections they agree with everyone else as well as right-handers agree with each other.
- **Two findings went against the original plan** and we followed the data: smoothing before derivative features *hurts* (turning it off scored higher in 4 of 5 folds with all sensors and 5 of 5 IMU-only), and deleting the upside-down subjects is worse than fixing them.
- **Two steps are correct but redundant for a tree model:** quaternion sign continuity and gravity removal no longer move the score once the 6D rotation and gravity-direction features exist. We report this rather than overclaim.

1. The task and what it demands from preprocessing

A wrist device records motion (accelerometer + orientation quaternion), five thermopiles (skin temperature) and five 8×8 time-of-flight (ToF) depth cameras while a person performs one of 18 gestures, 8 of which are body-focused repetitive behaviours (BFRBs). We must name the gesture from one short recording. Three facts about the test set drive every design decision:

- **One recording at a time.** Kaggle calls our code once per test sequence, so every step must work on a single recording; anything learned from all data (scalers) is fitted once and saved.
- **About half the test recordings have only the motion sensor.** The pipeline must work, and be evaluated, with thermopile and ToF missing.
- **Only sensor columns and demographics are available at test time.** Train-only columns (gesture, orientation, behaviour, phase) may be targets, never inputs.

2. What the data actually looks like (verified, not assumed)

Before writing any correction we measured the data (script `01_verify_physics.py`). Key facts:

Question	Answer on the real data	Used for
Size	574,945 rows, 8,151 recordings, 81 people, 18 gestures	folds, report
Recording length	median 59, 95th pct 127, max 700 rows	fixed length L = 128
Handedness coding	1 = right (71 people), 0 = left (10)	mirroring
Quaternion convention	rotating acceleration to the world frame gives 9.63 m/s² on the vertical (gravity)	gravity removal
Quaternion sign flips	**13,644 flips in 4,730 recordings (58%)**	continuity step
Rotation missing	entirely in 50 recordings, never partly	gravity fallback
Thermopile / ToF missing	entirely in 96 recordings (IMU-only); partly in 412 / 339; sensor 5 drops out most	gap filling, flags
ToF “no object” (-1)	58.6% of ToF pixels; valid range 0–249	ToF encoding
Where the gesture happens	gesture phase starts at 47% of the recording (median)	keep the end
Broken recordings	1 recording without gesture rows (flagged); no duplicate or missing counters	sanity checks

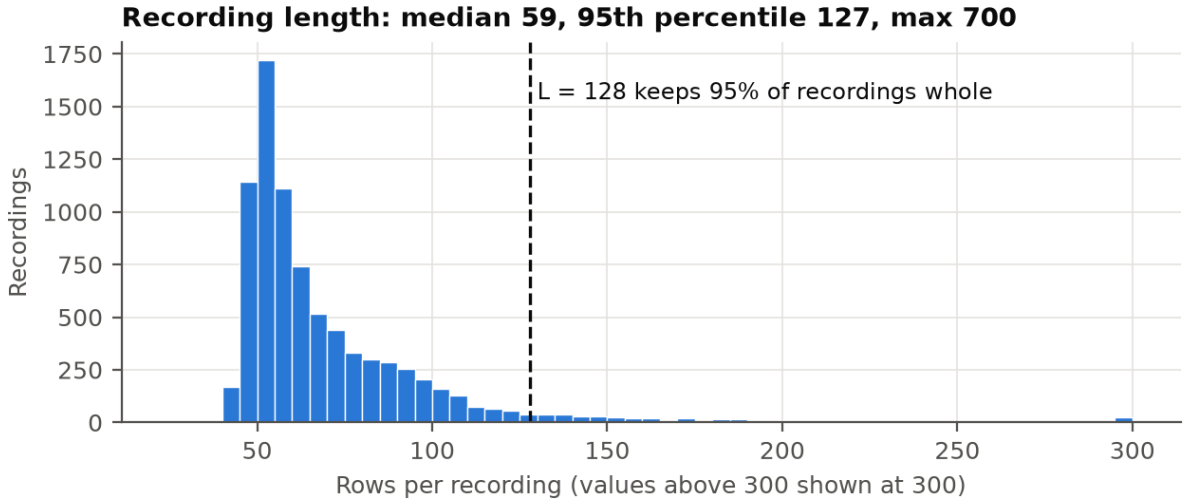


Figure 1. Recording lengths. Padding/truncating to L = 128 keeps 95% of recordings whole; longer ones keep their last 128 rows.

3. The pipeline, step by step: what we do and why

3.1 Cleaning (one recording at a time)

Step	What	Why
Order	sort rows by sequence_counter	everything after assumes time order; Kaggle may not send rows in order
Gaps	linear interpolation over the counter, nearest value at the ends; a sensor missing for the whole recording stays missing and is flagged	a filled gap is a better guess than zero; a dead sensor must not be faked
Quaternion sign	flip q to −q when consecutive rows point opposite ways, then normalise	q and −q are the same orientation; a flip looks like a sudden full twist (Figure 2)
Rotation gaps	interpolate on continuous quaternions (nlerp); identity only if rotation is absent for the whole recording	an identity row mid-recording would invent a huge twist
ToF encoding	proximity = 1 − distance/255; “-1, nothing in range” → 0 (far)	keeps the physical order: 1 touching … 0 far. The earlier cleaning mapped −1 to distance 0, i.e. “touching”
Robustness	strict mode audits training data; prediction mode repairs instead of crashing	one exception inside Kaggle fails the whole submission

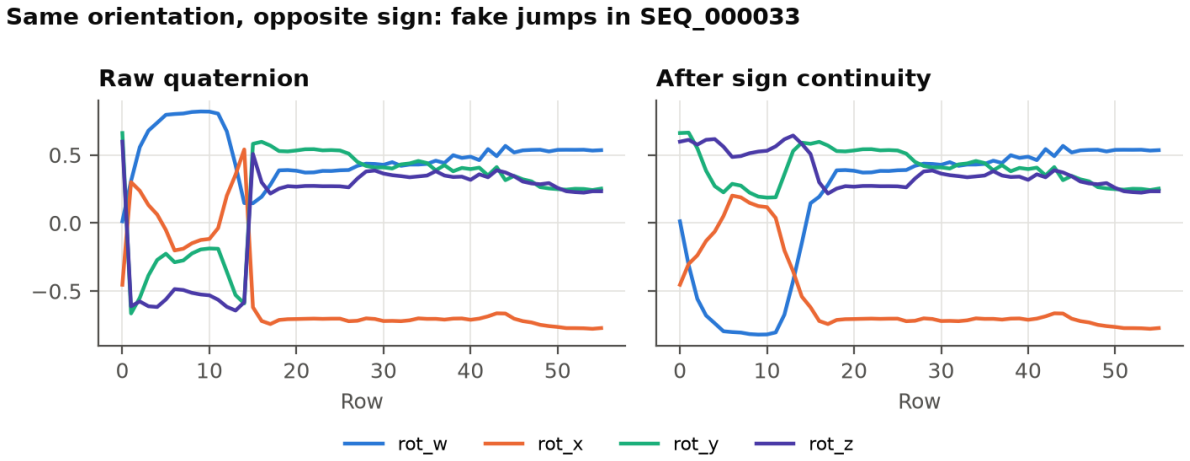


Figure 2. A real recording with quaternion sign flips. Left: the raw quaternion jumps although the wrist did not; right: after continuity the path is smooth.

3.2 Putting every recording into one device frame

The same gesture looks different to the watch if it is worn on the other wrist or upside down. Instead of trusting competition write-ups, we tested every candidate correction by asking: after the correction, how well does a subject’s average gravity direction per gesture agree with the right-handed population (mean cosine similarity, 1 = identical)? The method was first validated on synthetic data with a planted transform, where it recovered the planted axis, sensor swap and image flip exactly.

Left-handed: mirror which axis?	Agreement with right-handers
no mirroring	0.379
mirror x	**0.639** (right-handers among themselves: 0.637)
mirror y	-0.226
mirror z	-0.034

For left-handers, thermopile/ToF sensors 3 and 5 swap places (thermopile error 15.9 → 12.8, ToF error 0.046 → 0.030), and each 8×8 ToF image is flipped left-right on sensors 1, 2, 4 but up-down on 3 and 5, which are mounted sideways. All five sensors are corrected, not just the accelerometer.

Upside-down wearers. Exactly two subjects stand out (SUBJ_019262, SUBJ_045235); no other subject is flagged. A 180° rotation about the watch's z axis fixes them (agreement -0.203 → 0.622; rotating about x or y gives -0.013 / -0.406). Their sensors 2↔4 and 3↔5 swap and images rotate 180° (thermopile error 2.53 → 0.74).

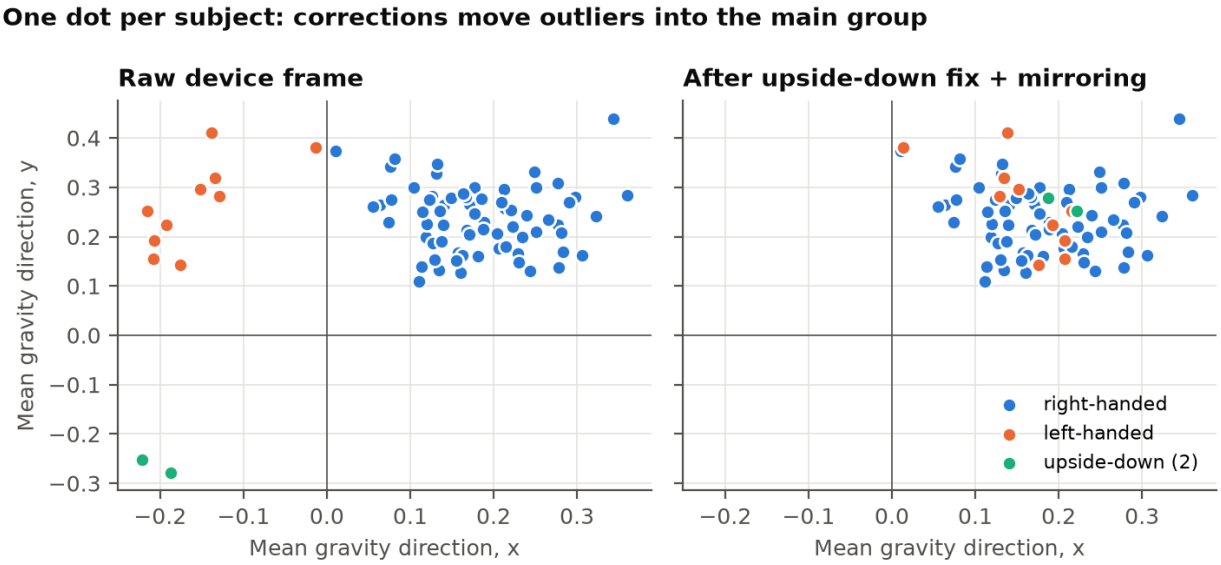


Figure 3. Each dot is one person's average gravity direction seen by the watch. Left-handers (orange) and the two upside-down wearers (green) sit apart in the raw data and join the main group after the corrections.

3.3 Features (computed per time step)

Group	Features	Why it should help
Motion	acceleration x/y/z and magnitude; **linear acceleration** (gravity removed via the quaternion, or a low-pass estimate if rotation is missing); vertical and horizontal motion; jerk; similarity between consecutive acceleration directions	separates the movement from posture; a scratch is fast and repetitive, a pinch is slow
Orientation	quaternion; **gravity direction** in the watch frame; continuous 6D rotation (no sign ambiguity); angular velocity and angular distance	hand pointing up (head) vs down (leg); twisting vs not
Thermopile	five temperatures, their mean, change per step	skin nearby is warm
ToF	per sensor: mean proximity, closest point, spread, fraction of pixels seeing something, change, and a **3×3 grid of regional averages**	how close the surface is and *where* in view it is
Context	flags for missing rotation / thermopile / ToF; position in the recording	lets one model handle every sensor situation

Total: 117 per-row features (for neural networks, padded to L = 128 with a mask, start-padded so the end is kept) and 795 per-recording statistics (mean, std, min, max, last value, and mean/std over the last 25%) for tree models. Values that do not exist (e.g. thermopile when the sensor is off) are NaN: tree models handle that natively; after scaling, neural networks get 0 plus the flag.

3.4 Labels, folds and scaling

- Labels (training targets only): 18-gesture, binary BFRB, the 9-class scoring label, per-row phase (transition / pause / gesture), orientation, and gesture × orientation × start-state (102 combinations).
- Five folds grouped by person and stratified by gesture, fixed seed, saved as one shared file: no person is in two folds, every fold has all 18 gestures and a BFRB share of 62.6–62.9%.
- Per-fold standardisation fitted on training rows only (no leakage), plus an all-data scaler for the final model.

4. Does each step help? Cumulative ablation

Same LightGBM settings, same five person-based folds, one change per row. Each model is trained on every training recording twice (all sensors, and with thermopile/ToF blanked) and scored on both versions of the held-out people.

Row	Adds	All sensors	Δ	IMU only	Δ
A	raw columns, missing → 0	0.750 ± 0.012		0.663 ± 0.015	
B	cleaning + ToF/thermopile summaries	0.764 ± 0.011	+0.014	0.670 ± 0.008	+0.008
C	statistics of the last 25% (keep the end)	0.790 ± 0.012	+0.026	0.707 ± 0.019	+0.037
D	upside-down subjects corrected	0.798 ± 0.013	+0.008	0.710 ± 0.020	+0.003
E	left-handers mirrored (all sensors)	0.807 ± 0.015	+0.009	0.709 ± 0.017	−0.001
F	gravity removal (+ fallback)	0.807 ± 0.015	0.000	0.722 ± 0.018	+0.012
G	engineered features, with smoothing	0.807 ± 0.016	0.000	0.730 ± 0.016	+0.009
H	**final: G without smoothing**	0.815 ± 0.013	+0.007	0.740 ± 0.016	+0.010

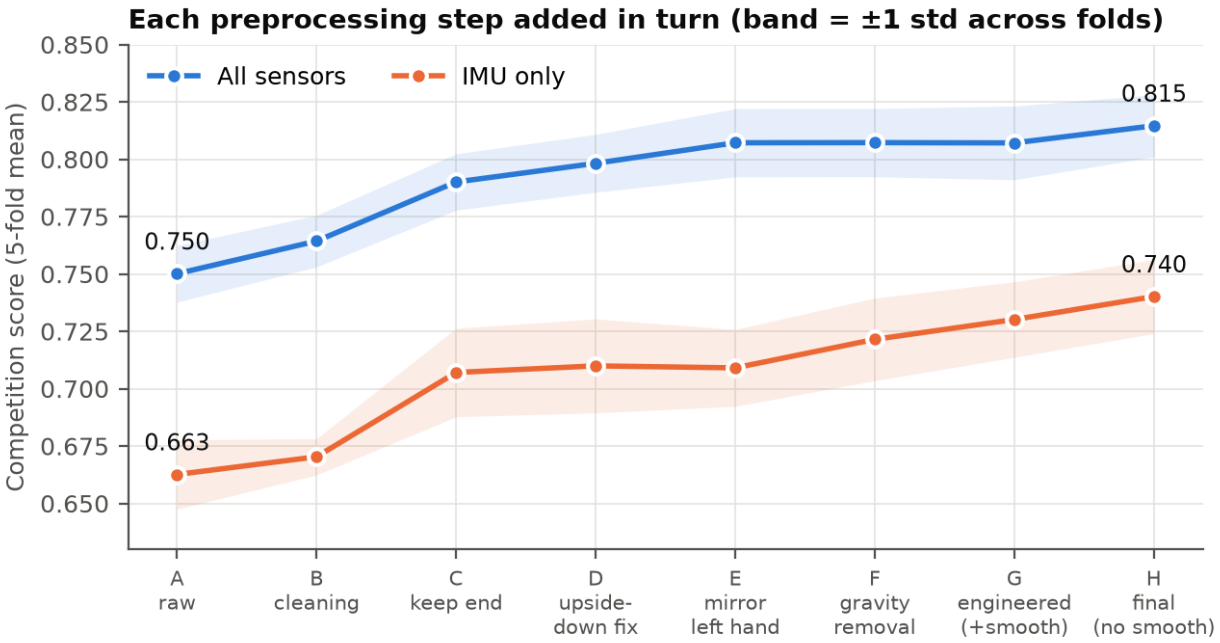


Figure 4. Score after each step. Shaded bands show ±1 standard deviation across the five folds.

5. Does each step still matter with everything else in? Leave-one-out

Starting from the full pipeline G, we removed one step at a time. Because the fold-to-fold spread (≈ ±0.015) is larger than several differences, we also count in how many of the five folds G beats the variant: a step that wins 5/5 is reliable even if its average gain is small.

Variant of G	All sensors	Δ vs G	Folds G wins	IMU only	Δ vs G	Verdict
remove ToF 3×3 regions	0.794	−0.013	5/5	0.733	+0.003	keep: largest single step
ToF −1 as “touching” (earlier cleaning)	0.796	−0.011	5/5	0.734	+0.004	keep “far”
remove left-hand mirroring	0.797	−0.010	5/5	0.727	−0.003	keep
delete upside-down subjects instead of fixing	0.799	−0.008	3/5	0.729	−0.002	fix, don’t delete
remove upside-down fix	0.801	−0.006	4/5	0.730	0.000	keep
remove quaternion continuity	0.807	0.000	1/5	0.729	−0.001	no effect here
remove gravity removal	0.809	+0.002	3/5	0.734	+0.004	redundant for trees
remove smoothing	0.815	+0.007	0/5	0.740	+0.010	**smoothing hurts → off**

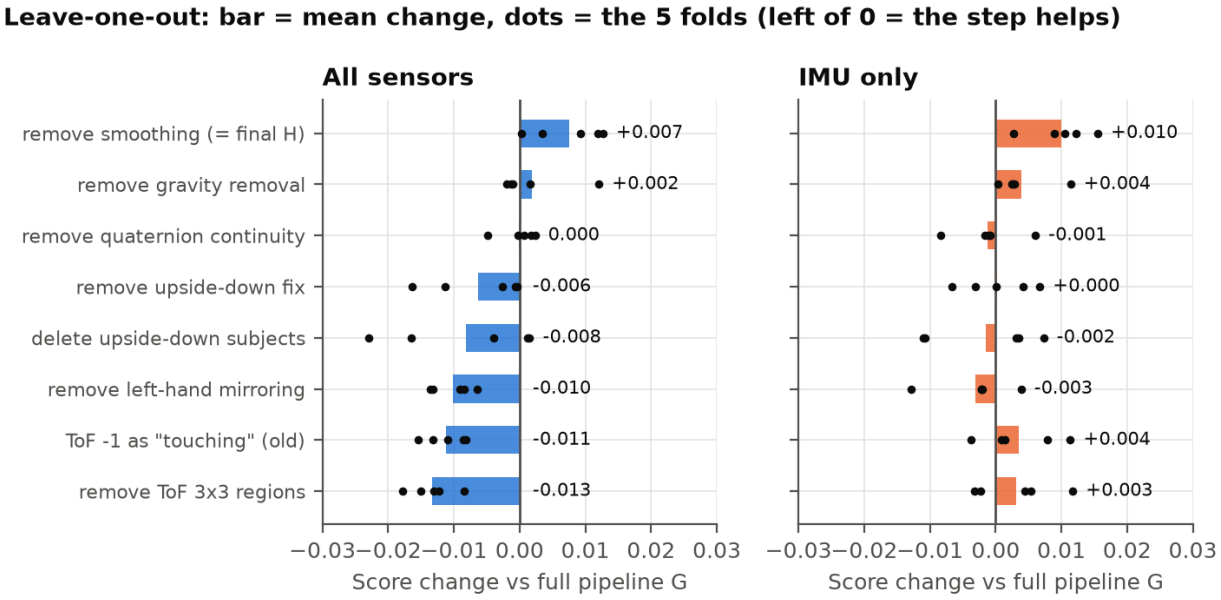


Figure 5. Change in score when one step is removed from G. Bars: mean over folds; dots: each fold. Bars left of zero mean the step was helping.

6. Honest reading of the results

What clearly works. ToF handling is the largest contributor once everything else is in: without the 3×3 regions the all-sensor score drops by 0.013, and the earlier "−1 = touching" encoding costs 0.011; both lose in all five folds. Left-hand mirroring (−0.010, 5/5 folds) and the upside-down fix (−0.006) also hold up. Deleting the upside-down subjects instead of fixing them is worse than fixing them, and even worse than leaving them alone: their data is valuable once corrected.

Smoothing hurts, so we turned it off. The original plan smoothed motion before computing jerk and angular velocity. Removing it raised the score from 0.807 to 0.815 (all sensors) and 0.730 to 0.740 (IMU only), better in 4 of 5 folds (all sensors; the remaining fold is a tie) and 5 of 5 (IMU only). At about 10 readings per second a gesture lasts only a few rows, so even light smoothing blurs the quick changes that distinguish, for example, scratching from pinching.

Correct but redundant for a tree model. Quaternion continuity mattered on its own (it is part of row B), but once the 6D rotation is used, which has no sign ambiguity by construction, removing continuity changes nothing. Likewise gravity removal clearly helped when added (row F: +0.012 IMU only), but in the full pipeline the gravity-direction and 6D-rotation features let the trees recover the same information. Both steps stay: they are physically right and cheap, and neural networks, which do not split on raw columns the way trees do, may still benefit. That is a test for the modelling team.

Limits of this evidence. The yardstick is one tree model; neural networks may weigh steps differently. Differences under about 0.005 are within fold noise. Augmentations (section 8b of the guide) were implemented but cannot be measured without training a sequence model, so they are left to the modellers.

7. How we know the preprocessing is correct

Check	Result
Unit tests (synthetic recordings built from known physics)	44 / 44 pass (33 new + 11 from the earlier cleaning)
Physics properties	mirroring, the upside-down fix and rotation augmentations keep gravity consistent with the quaternion to 10^{-9}
Synthetic dataset with planted transforms	verification script recovered the planted mirror axis, sensor swap, image flip and both upside-down subjects exactly
Train ↔ Kaggle parity	377 real recordings sent through the Kaggle path (polars, shuffled rows, no label columns): max difference 0.0; real test sequences 2/2
Output quality checks	22 / 22 pass, e.g. no subject appears in two folds; rotation features NaN exactly when rotation is missing; sensor columns NaN only for whole-recording dropouts (gaps filled); no train-only column leaks into features; gravity direction is a unit vector
Build	8,151 recordings, 574,945 rows, 0 data issues

8. Compared with the earlier cleaning

Earlier cleaning	Now	Measured effect
ToF −1 → 0 ("touching")	proximity, −1 → far	+0.011 all sensors, 5/5 folds
missing rotation rows → identity	interpolated; identity only if absent all recording	no partial gaps exist in train; protects test
no quaternion continuity	continuity	58% of recordings had flips
raised errors on bad input	repairs at prediction time	cannot crash the Kaggle run
no orientation fixes, features, labels, folds	all added	+0.065 / +0.078 overall

The earlier work (kept in `legacy/`) contributed the strict input validation, the sequence-local design and the original test suite, all of which carried over.

9. What the modelling team receives

- `processed/`: per-row features, per-recording tree features (all sensors and IMU-only), labels, the shared fold file, per-fold scalers, and cleaned signals for on-the-fly augmentation.
- `preprocess(sequence, demographics)`: call it inside the Kaggle `predict()`; it accepts the polars frames Kaggle sends and returns the same features as training.
- `preprocessing/augmentation.py`: crop, time-stretch, wrist/heading rotation, magnitude scaling, jitter, sensor dropout and mixup, applied to raw signals so physics features stay consistent.

10. Sources (ideas only; no code copied)

CMI competition top-solution write-ups (1st–15th place summaries; 7th-place slides on handedness, upside-down subjects and gravity removal; 6th-place note on upside-down detection); silver-medal solution README (angular velocity, ToF aggregation); Zhou et al. 2019, *On the Continuity of Rotation Representations in Neural Networks* (6D rotation); Um et al. 2017, *Data Augmentation of Wearable Sensor Data*. All choices were re-verified on our data as described above.